QUICK DIAGNOSTIC CHART [SKYACTIV-D 1.5]

id0103q2052800

×: Applicable
—: Not applicable

Troubleshooting item		Intake air system								
		Intake air system part		Air cleaner			Intake shutter valve		Compressor	Electric water pump
		Air suction/ Leakage/ Perforation	Clogging	Deformation	Air cleaner element restricted	Non-genuine part installation	Does not fully close (Does not conform to target value)	Clogging, sticking	Improper operation	Improper operation
1	Blown fuses	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	×	×	—	×	—	×	×	×	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	×
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/ jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—

Troubleshooting item		Intake air system								
		Intake air system part		Air cleaner			Intake shutter valve		Compressor	Electric water pump
		Air suction/ Leakage / Perforation	Clogging	Deformation	Air cleaner element restricted	Non-genuine part installation	Does not fully close (Does not conform to target value)	Clogging, sticking	Improper operation	Improper operation
17	Lack/loss of power-acceleration/cruise	—	—	—	—	×	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	×	—	—	—	×
20	Emission compliance	—	—	—	—	×	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—
26	Engine noise	×	—	×	—	×	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)								

×: Applicable
—: Not applicable

Troubleshooting item		Intake air system				Exhaust system				
		Water-cooled charge air cooler		Improper installation of turbocharger	Interference with housing of turbo rotational mechanism	Each exhaust system part		Variable turbine geometry turbocharger	Turbocharger actuator	Turbocharger solenoid valve
		Leakage	Clogging			Leakage	Clogging (mostly in oxidation catalytic converter for diesel, diesel particulate filter)	Improper operation	Improper operation	Improper operation
1	Blown fuses	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	×	×	—	—	—

Troubleshooting item		Intake air system				Exhaust system				
		Water-cooled charge air cooler		Improper installation of turbocharger	Interference with housing of turbo rotational mechanism	Each exhaust system part		Variable turbine geometry turbocharger	Turbocharger actuator	Turbocharger solenoid valve
		Leakage	Clogging			Leakage	Clogging (mostly in oxidation catalytic converter for diesel, diesel particulate filter)	Improper operation	Improper operation	Improper operation
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	×	×	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/ jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—

Troubleshooting item		Intake air system				Exhaust system				
		Water-cooled charge air cooler		Improper installation of turbocharger	Interference with housing of turbo rotational mechanism	Each exhaust system part		Variable turbine geometry turbocharger	Turbocharger actuator	Turbocharger solenoid valve
		Leakage	Clogging			Leakage	Clogging (mostly in oxidation catalytic converter for diesel, diesel particulate filter)	Improper operation	Improper operation	Improper operation
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	×	—	—
19	Poor fuel economy	×	×	—	—	—	—	—	×	×
20	Emission compliance	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	×	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	×	×	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)								

×: Applicable
—: Not applicable

Troubleshooting item		Exhaust system								
		HP-EGR control valve	LP-EGR control valve	Exhaust shutter valve	LP-EGR cooler		HP-EGR control valve	LP-EGR control valve	EGR filter	Diesel particulate filter
		Improper operation	Improper operation	Improper operation	Leakage (gas)	Clogging	Clogging	Clogging	Clogging	Loss
1	Blown fuses	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	×	—	—	×	×	—	—	×	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—

Troubleshooting item		Exhaust system								
		HP-EGR control valve	LP-EGR control valve	Exhaust shutter valve	LP-EGR cooler		HP-EGR control valve	LP-EGR control valve	EGR filter	Diesel particulate filter
		Improper operation	Improper operation	Improper operation	Leakage (gas)	Clogging	Clogging	Clogging	Clogging	Loss
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	×	×	—	—	—	—
19	Poor fuel economy	—	×	×	—	—	—	—	×	—
20	Emission compliance	—	—	—	—	×	×	—	×	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	×
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—

Troubleshooting item		Exhaust system								
		HP-EGR control valve	LP-EGR control valve	Exhaust shutter valve	LP-EGR cooler		HP-EGR control valve	LP-EGR control valve	EGR filter	Diesel particulate filter
		Improper operation	Improper operation	Improper operation	Leakage (gas)	Clogging	Clogging	Clogging	Clogging	Loss
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)								

×: Applicable
—: Not applicable

Troubleshooting item		Fuel system								
		Fuel line	Fuel filter		Fuel warmer	Supply pump	Suction control valve		Fuel pressure relief valve	
		Leakage	Leakage	Clogging	Improper operation	Poor discharge amount	Low adjusted pressure value	High adjusted pressure value	Low adjusted pressure value	High adjusted pressure value
1	Blown fuses	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	×	×	×	×	×	×	×	×	×
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—

Troubleshooting item		Fuel system								
		Fuel line	Fuel filter		Fuel warmer	Supply pump	Suction control valve		Fuel pressure relief valve	
		Leakage	Leakage	Clogging	Improper operation	Poor discharge amount	Low adjusted pressure value	High adjusted pressure value	Low adjusted pressure value	High adjusted pressure value
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item			(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)							

×: Applicable
—: Not applicable

Troubleshooting item		Fuel system							
		Fuel system/ Fuel injection system malfunction (such as leakage)	Fuel injector						
			Open circuit in drive circuit	Short circuit in drive circuit	Leakage	Restricted or clogged	Always open	Always close	QR code mismatch
1	Blown fuses	—	—	—	—	—	—	—	—

Troubleshooting item		Fuel system							
		Fuel system/ Fuel injection system malfunction (such as leakage)	Fuel injector						
			Open circuit in drive circuit	Short circuit in drive circuit	Leakage	Restricted or clogged	Always open	Always close	QR code mismatch
2	Check engine light illuminates	—	×	×	×	×	×	×	×
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—
10	Hard to start/long crank/erratic start/erratic crank	×	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	×	—	—	—	—	—	—	—
12	Cranks normally but will not start	×	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	×	—	—	—	—	—	—	—
14	Fast idle/runs on	×	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	×	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	×	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	×	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	×	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—
20	Emission compliance	×	—	—	—	—	—	—	—
21	High oil consumption/leakage	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	×	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—

Troubleshooting item		Fuel system							
		Fuel system/ Fuel injection system malfunction (such as leakage)	Fuel injector						
			Open circuit in drive circuit	Short circuit in drive circuit	Leakage	Restricted or clogged	Always open	Always close	QR code mismatch
27	Vibration concerns (engine)	×	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)							

×: Applicable
—: Not applicable

Troubleshooting item		Fuel system			Control system					
		Fuel injector	Fuel		PCM				Main relay	
		Injection amount correction miss-match	Low cetane value	Inferior constituent	No initiation	No configuration	No learning	Open circuit in back-up power supply circuit of PCM	Always off	Always on
1	Blown fuses	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	×	—	×	—	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	×	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	×	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	×	—	—	—	×	—

Troubleshooting item		Fuel system			Control system					
		Fuel injector	Fuel		PCM				Main relay	
		Injection amount correction miss-match	Low cetane value	Inferior constituent	No initiation	No configuration	No learning	Open circuit in back-up power supply circuit of PCM	Always off	Always on
10	Hard to start/long crank/ erratic start/erratic crank	—	×	×	—	—	×	—	—	—
11	Engine stalls-after start/at idle	—	×	×	—	—	×	—	—	—
12	Cranks normally but will not start	—	×	×	—	—	×	—	—	—
13	Engine runs rough/rolling idle	×	×	×	—	—	×	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	×	×	—	—	×	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	×	×	—	—	×	—	—	—
17	Lack/loss of power-acceleration/cruise	—	×	×	—	—	×	—	—	—
18	Knocking/pinging-acceleration/cruise	×	—	×	—	—	—	—	—	—
19	Poor fuel economy	—	×	×	—	—	×	—	—	—
20	Emission compliance	×	×	×	—	—	×	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	×	—	—	—
27	Vibration concerns (engine)	×	×	×	—	—	×	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)								

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM input signal)											
		MAF sensor			Intake shutter valve position sensor		APP sensor		CKP sensor			CMP sensor	
		High input	Low input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	False signal output (no noise, pulse)	Open or short circuit in signal circuit	Engine speed signal	False signal output (no noise, pulse)	Open or short circuit in signal circuit
1	Blown fuses	—	—	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	×	×	×	×	×	×	×	×	×	—	×	×
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	×	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)											
		MAF sensor			Intake shutter valve position sensor		APP sensor		CKP sensor			CMP sensor	
		High input	Low input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	False signal output (no noise, pulse)	Open or short circuit in signal circuit	Engine speed signal	False signal output (no noise, pulse)	Open or short circuit in signal circuit
17	Lack/loss of power-acceleration/cruise	—	—	—	×	×	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)											

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM input signal)											
		ECT sensor			IAT sensor No.1			Boost air temperature sensor			IAT sensor No.2		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
1	Blown fuses	—	—	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	×	—	—	×	—	—	×	—	—	×
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)											
		ECT sensor			IAT sensor No.1			Boost air temperature sensor			IAT sensor No.2		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	×	×	—	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	×	×	—	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	×	×	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	×	×	—	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	×	×	—	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	×	×	—	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	×	×	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	×	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	×	—	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)											
		ECT sensor			IAT sensor No.1			Boost air temperature sensor			IAT sensor No.2		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	×	×	—	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)											

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM input signal)								
		MAP sensor			Fuel temperature sensor			Fuel pressure sensor		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
1	Blown fuses	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	×	—	—	×	×	—	×
3	Diesel particulate filter indicator light illuminated frequently/ message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/ message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/erratic start/erratic crank	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)								
		MAP sensor			Fuel temperature sensor			Fuel pressure sensor		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	×	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/ cruise	×	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/ cruise	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—
21	High oil consumption/leakage	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	×	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)								

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM input signal)											
		Exhaust gas temperature sensor No.1			Exhaust gas pressure sensor No. 1			Engine oil temperature sensor			Engine oil pressure sensor		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
1	Blown fuses	—	—	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	×	—	—	×	—	—	×	—	×	×
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)											
		Exhaust gas temperature sensor No.1			Exhaust gas pressure sensor No. 1			Engine oil temperature sensor			Engine oil pressure sensor		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	×	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	×	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	×	—	—	×	×	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)											
		Exhaust gas temperature sensor No.1			Exhaust gas pressure sensor No. 1			Engine oil temperature sensor			Engine oil pressure sensor		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	×	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)											

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM input signal)											
		Exhaust gas temperature sensor No.2			Exhaust gas temperature sensor No.3			Exhaust gas temperature sensor No.4			Exhaust gas pressure sensor No.2		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
1	Blown fuses	—	—	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	×	×	×	—	×	×	×	×	×	×	×	×
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	×	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	×	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)											
		Exhaust gas temperature sensor No.2			Exhaust gas temperature sensor No.3			Exhaust gas temperature sensor No.4			Exhaust gas pressure sensor No.2		
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/ jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item	Control system (PCM input signal)											
	Exhaust gas temperature sensor No.2			Exhaust gas temperature sensor No.3			Exhaust gas temperature sensor No.4			Exhaust gas pressure sensor No.2		
	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit
i-stop related troubleshooting item	(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)											

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM input signal)													
		A/F sensor			BARO sensor (built-intoPCM)			HP-EGR control valve position sensor		LP-EGR control valve position sensor		Exhaust shutter valve position sensor		Speed sensor assy (turbocharger with variable turbine geometry)	
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit
1	Blown fuses	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	×	×	×	—	—	×	×	×	×	×	×	×	—	×
3	Diesel particulate filter indicator light illuminated frequently/ message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)													
		A/F sensor			BARO sensor (built-intoPCM)			HP-EGR control valve position sensor		LP-EGR control valve position sensor		Exhaust shutter valve position sensor		Speed sensor assy (turbocharger with variable turbine geometry)	
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit
6	Master warning light illuminated/ message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/ message related to charging system malfunction indicated in display	—	—	—	×	—	×	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/ message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/erratic start/ erratic crank	—	—	—	—	—	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	×	×	×	×	×	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—	—	—	—	—	—
16	Engine stalls/ quits, engine runs rough, misses, buck/jerk, hesitation/ stumble, surges	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)												
		A/F sensor			BARO sensor (built-intoPCM)			HP-EGR control valve position sensor		LP-EGR control valve position sensor		Exhaust shutter valve position sensor		Speed sensor assy (turbocharger with variable turbine geometry)	
		High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit	Erratic signal input	Open or short circuit in signal circuit
17	Lack/loss of power-acceleration/ cruise	×	×	×	×	×	—	×	×	×	×	×	×	×	×
18	Knocking/pinging-acceleration/ cruise	—	—	—	—	—	—	—	—	×	×	—	—	×	×
19	Poor fuel economy	—	—	—	—	—	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	×	×	×	—	—	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)													

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM input signal)												
		Actuator position sensor (turbocharger with variable turbine geometry)		Generator output current signal			Refrigerant pressure sensor			Check connector	CPP switch (MTX)		Starter interlock switch (MTX)	
		Erratic signal input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	Open circuit	Always on	Always off	Always on	Always off
1	Blown fuses	—	—	—	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	×	×	—	—	—	—	—	—	—	×	×	—	—
3	Diesel particulate filter indicator light illuminated frequently/ message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	×	—	×	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/ at idle	—	—	—	—	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/ rolling idle	—	—	—	—	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)												
		Actuator position sensor (turbocharger with variable turbine geometry)		Generator output current signal			Refrigerant pressure sensor			Check connector	CPP switch (MTX)		Starter interlock switch (MTX)	
		Erratic signal input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	High input	Low input	Open or short circuit in signal circuit	Open circuit	Always on	Always off	Always on	Always off
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	×	×	—	—	—	—	×	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	×	×	—	—	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	×	×	×	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)												

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM input signal)								Control system (PCM output signal)			
		Brake switch (No.1 signal)		Brake switch (No.2 signal)		Neutral switch (MTX)		Back-up light switch (MTX)		Fan control module control circuit			
		Always on	Always off	Always on	Always off	Always on	Always off	Always on	Always off	Open circuit in power supply circuit	Control signal circuit always on	Control signal circuit always off	Open circuit in ground circuit
1	Blown fuses	—	—	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	×	×	×	×	×	×	—	—	—	—	—	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/ message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/ at idle	—	—	—	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/ rolling idle	—	—	—	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM input signal)								Control system (PCM output signal)			
		Brake switch (No.1 signal)		Brake switch (No.2 signal)		Neutral switch (MTX)		Back-up light switch (MTX)		Fan control module control circuit			
		Always on	Always off	Always on	Always off	Always on	Always off	Always on	Always off	Open circuit in power supply circuit	Control signal circuit always on	Control signal circuit always off	Open circuit in ground circuit
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—	×	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	×	—	×	×
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	×	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)											

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM output signal)									
		A/C relay circuit			Generator control signal	Starter relay circuit			Blow-by heater relay circuit		
		Open circuit in circuit on coil side	Circuit on switch side constantly on	Circuit on switch side constantly off	Open or short circuit	Open circuit in circuit on coil side	Circuit on switch side constantly on	Circuit on switch side constantly off	Open circuit in circuit on coil side	Circuit on switch side constantly on	Circuit on switch side constantly off
1	Blown fuses	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	—	—	—	—	—	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM output signal)									
		A/C relay circuit			Generator control signal	Starter relay circuit			Blow-by heater relay circuit		
		Open circuit in circuit on coil side	Circuit on switch side constantly on	Circuit on switch side constantly off	Open or short circuit	Open circuit in circuit on coil side	Circuit on switch side constantly on	Circuit on switch side constantly off	Open circuit in circuit on coil side	Circuit on switch side constantly on	Circuit on switch side constantly off
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	×	×	×
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	×	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	×	—	×	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	×	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/ jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	×	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	×	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	×	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM output signal)									
		A/C relay circuit			Generator control signal	Starter relay circuit			Blow-by heater relay circuit		
		Open circuit in circuit on coil side	Circuit on switch side constantly on	Circuit on switch side constantly off	Open or short circuit	Open circuit in circuit on coil side	Circuit on switch side constantly on	Circuit on switch side constantly off	Open circuit in circuit on coil side	Circuit on switch side constantly on	Circuit on switch side constantly off
26	Engine noise	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	×	—	×	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	×	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)									

×: Applicable
—: Not applicable

Troubleshooting item		Control system (PCM output signal)						Mechanical				
		Glow control module control signal				A/F sensor heater		Each engine internal part				
		Open circuit in power supply circuit	Control signal circuit constantly energizes (FET malfunction)	Control signal circuit always off	Open circuit in ground circuit	Always on	Always off	Large mechanical resistance	Deviation in valve timing	High compression pressure	No compression in specific cylinder	No compression in all cylinders
1	Blown fuses	—	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	×	×	×	×	×	×	—	—	—	—	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM output signal)						Mechanical				
		Glow control module control signal				A/F sensor heater		Each engine internal part				
		Open circuit in power supply circuit	Control signal circuit constantly energizes (FET malfunction)	Control signal circuit always off	Open circuit in ground circuit	Always on	Always off	Large mechanical resistance	Deviation in valve timing	High compression pressure	No compression in specific cylinder	No compression in all cylinders
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	×	×	×	×	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	×	×	×	×	—
12	Cranks normally but will not start	—	—	—	—	—	—	×	×	×	×	×
13	Engine runs rough/rolling idle	—	—	—	—	—	—	×	×	×	×	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	×	×	×	×	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	×	×	×	×	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	×	×	×	×	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	×	×	×	×	—
19	Poor fuel economy	—	—	—	—	—	—	×	×	×	×	—
20	Emission compliance	—	—	—	—	—	—	×	×	×	×	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	×	×	×	—
27	Vibration concerns (engine)	—	—	—	—	—	—	×	×	×	×	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Control system (PCM output signal)						Mechanical				
		Glow control module control signal				A/F sensor heater		Each engine internal part				
		Open circuit in power supply circuit	Control signal circuit constantly energizes (FET malfunction)	Control signal circuit always off	Open circuit in ground circuit	Always on	Always off	Large mechanical resistance	Deviation in valve timing	High compression pressure	No compression in specific cylinder	No compression in all cylinders
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)										

×: Applicable
—: Not applicable

Troubleshooting item		Mechanical									
		Each engine internal part				Blow-by heater		Engine oil solenoid valve			
		Bad compression in specific cylinder	Bad compression in all cylinders	Oil working up or down	Piston ring malfunction	Always on	Always off	Open circuit in drive circuit	Short circuit in drive circuit	Valve stuck open	Valve stuck close
1	Blown fuses	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	—	—	×	×	—	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	×	×	—	—	×	×
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Mechanical									
		Each engine internal part				Blow-by heater		Engine oil solenoid valve			
		Bad compression in specific cylinder	Bad compression in all cylinders	Oil working up or down	Piston ring malfunction	Always on	Always off	Open circuit in drive circuit	Short circuit in drive circuit	Valve stuck open	Valve stuck close
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	×	×	×	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	×	×	×	—	—	—	—	—	—	—
12	Cranks normally but will not start	×	×	×	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	×	×	×	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	×	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	×	×	×	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/ jerk, hesitation/stumble, surges	×	×	×	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	×	×	×	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	×	×	×	—	—	—	—	—	—	—
19	Poor fuel economy	×	×	×	—	—	—	—	—	—	—
20	Emission compliance	×	×	×	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	×	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—
26	Engine noise	×	×	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	×	×	×	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)									

×: Applicable
—: Not applicable

Troubleshooting item		Mechanical	Lubrication system				Cooling system			
		Belts for engine accessories	Engine oil				Engine coolant passage		Coolant control valve	Cooling fan motor
		Slippage	Inferior constituent	Leakage	Clogging	Improper level	Restricted or clogged	Leakage	Improper operation	Improper operation
1	Blown fuses	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	—	—	—	×	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	×	—	—	—	—	—	×	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	×	—	—	—	—	—	×	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	×	×	×	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/ jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	×	×
19	Poor fuel economy	—	—	—	—	—	—	—	×	—
20	Emission compliance	—	—	—	—	—	—	—	—	—

Troubleshooting item		Mechanical	Lubrication system				Cooling system			
		Belts for engine accessories	Engine oil				Engine coolant passage		Coolant control valve	Cooling fan motor
		Slippage	Inferior constituent	Leakage	Clogging	Improper level	Restricted or clogged	Leakage	Improper operation	Improper operation
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	×	×	×	×
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	×	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—
26	Engine noise	×	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)								

×: Applicable
—: Not applicable

Troubleshooting item		Charging system								
		Battery		Generator			Starter		Glow plug	
		Does not charge	No discharge (deterioration)	Low power generation voltage	No power generation	High power generation	Open/ short circuit in drive circuit	Internal malfunction	Always off	Always on
1	Blown fuses	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	—	—	—	×	×
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—

Troubleshooting item		Charging system								
		Battery		Generator			Starter		Glow plug	
		Does not charge	No discharge (deterioration)	Low power generation voltage	No power generation	High power generation	Open/ short circuit in drive circuit	Internal malfunction	Always off	Always on
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	×	×	×	×	×	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—
9	Will not crank	×	×	—	—	—	×	×	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/ jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)								

×: Applicable
—: Not applicable

Troubleshooting item		Emission system	ATX		ATX			Other			
		Catalytic converter (diesel particulate filter)	ATF		Automatic transaxle			Engine mount	Instrument cluster	Brake	Tire
		Deterioration or loss	Oil cooler restriction	Lack of ATF	TCC non-operation	Constant lockup	Abnormal shift point	Looseness	Improper operation	Drag	Low air pressure
1	Blown fuses	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	×	—	—	—	—	—	×	—	—
3	Diesel particulate filter indicator light illuminated frequently/ message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/ message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/ message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/ message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	×	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/erratic start/ erratic crank	—	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	×	×	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/ rolling idle	—	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		Emission system	ATX		ATX			Other			
		Catalytic converter (diesel particulate filter)	ATF		Automatic transaxle			Engine mount	Instrument cluster	Brake	Tire
		Deterioration or loss	Oil cooler restriction	Lack of ATF	TCC non-operation	Constant lockup	Abnormal shift point	Loosenness	Improper operation	Drag	Low air pressure
15	Low idle/stalls during deceleration	—	—	—	—	×	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	×	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	×	—	×	—	—	×	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	×	—	×	—	—	×	×
20	Emission compliance	×	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	×	×	×	×	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	×	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)									

×: Applicable
—: Not applicable

Troubleshooting item		Other								
		Vehicle electric devices			A/C compressor	Immobilizer system		Advanced keyless entry system	Climate control unit	Magnetic clutch
		Remote transmitter left in vehicle cabin	Large amount of parasitic draw	Large power consumption	Large mechanical resistance	Improper operation	Unprogrammed remote transmitter	Improper operation	Improper operation	Improper operation
1	Blown fuses	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	—	—	—	—	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	×	×	×	×	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	×	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	×	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/ jerk, hesitation/stumble, surges	—	—	—	×	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—

Troubleshooting item		Other								
		Vehicle electric devices			A/C compressor	Immobilizer system		Advanced keyless entry system	Climate control unit	Magnetic clutch
		Remote transmitter left in vehicle cabin	Large amount of parasitic draw	Large power consumption	Large mechanical resistance	Improper operation	Unprogrammed remote transmitter	Improper operation	Improper operation	Improper operation
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	×	×
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)								

×: Applicable
—: Not applicable

Troubleshooting item		i-stop										
		i-stop OFF signal malfunction	Bonnet latch switch		Door latch switch or trunk lid latch switch (4SD)/ liftgate latch switch (5HB)		False detection of driver-side buckle switch		Instrument cluster malfunction	Number of times i-stop function operates exceeds assured operation amount of components	Ambient temperature sensor	Cabin temperature sensor
			Stuck open	Stuck close	Stuck open	Stuck close	Stuck off	Stuck on			Erratic signal input	Erratic signal input
1	Blown fuses	—	—	—	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	—	—	—	—	—	—	—
3	Diesel particulate filter indicator light illuminated frequently/ message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		i-stop										
		i-stop OFF signal malfunction	Bonnet latch switch		Door latch switch or trunk lid latch switch (4SD)/ liftgate latch switch (5HB)		False detection of driver-side buckle switch		Instrument cluster malfunction	Number of times i-stop function operates exceeds assured operation amount of components	Ambient temperature sensor	Cabin temperature sensor
			Stuck open	Stuck close	Stuck open	Stuck close	Stuck off	Stuck on			Erratic signal input	Erratic signal input
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	×	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—	—	—	—
11	Engine stalls-after start/ at idle	—	—	—	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—	—	—	—
13	Engine runs rough/ rolling idle	—	—	—	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—	—	—	—

Troubleshooting item		i-stop										
		i-stop OFF signal malfunction	Bonnet latch switch		Door latch switch or trunk lid latch switch (4SD)/ liftgate latch switch (5HB)		False detection of driver-side buckle switch		Instrument cluster malfunction	Number of times i-stop function operates exceeds assured operation amount of components	Ambient temperature sensor	Cabin temperature sensor
			Stuck open	Stuck close	Stuck open	Stuck close	Stuck off	Stuck on			Erratic signal input	Erratic signal input
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)										

×: Applicable
—: Not applicable

Troubleshooting item		i-stop							
		Solar radiation sensor	Driver-side air mix actuator malfunction	Driver-side air mix actuator position sensor	Steering angle sensor	Defroster switch malfunction	Low power brake unit load	Power brake unit vacuum sensor	Battery management system malfunction
		Erratic signal input		Erratic signal input	Erratic signal input			Erratic signal input	
1	Blown fuses	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	—	—	—	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/ message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—

Troubleshooting item		i-stop							
		Solar radiation sensor	Driver-side air mix actuator malfunction	Driver-side air mix actuator position sensor	Steering angle sensor	Defroster switch malfunction	Low power brake unit load	Power brake unit vacuum sensor	Battery management system malfunction
		Erratic signal input		Erratic signal input	Erratic signal input			Erratic signal input	
8	Engine oil warning light illuminated/message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—
10	Hard to start/long crank/ erratic start/erratic crank	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—
13	Engine runs rough/rolling idle	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	×
20	Emission compliance	—	—	—	—	—	—	—	—
21	High oil consumption/ leakage	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)							

×: Applicable
—: Not applicable

Troubleshooting item		i-stop							
		Brake fluid pressure sensor (built-into DSC HU/CM)	M position switch malfunction (ATX)	Low-G (XY) sensor (built-into SAS control module)	TFT sensor (ATX)	Electric AT oil pump malfunction (ATX)	DSC HU/CM malfunction	Generator drive belt wear	Ignition switch signal (PCM terminal 2H)
			Erratic signal input	Erratic signal input	Erratic signal input				Stuck on
1	Blown fuses	—	—	—	—	—	—	—	—
2	Check engine light illuminates	—	—	—	—	—	—	—	—
3	Diesel particulate filter indicator light illuminated frequently/message related to accumulation condition in diesel particulate filter indicated in display	—	—	—	—	—	—	—	—
4	Diesel particulate filter indicator light flashes/message related diesel particulate filter malfunction indicated in display	—	—	—	—	—	—	—	—
5	Wrench warning indicator light illuminated or flashes	—	—	—	—	—	—	—	—
6	Master warning light illuminated/message related to vehicle system malfunction indicated in display	—	—	—	—	—	—	—	—
7	Charging system warning light illuminated/message related to charging system malfunction indicated in display	—	—	—	—	—	—	—	—
8	Engine oil warning light illuminated/ message related to engine hydraulic pressure indicated in display	—	—	—	—	—	—	—	—
9	Will not crank	—	—	—	—	—	—	—	—
10	Hard to start/long crank/erratic start/ erratic crank	—	—	—	—	—	—	—	—
11	Engine stalls-after start/at idle	—	—	—	—	—	—	—	—
12	Cranks normally but will not start	—	—	—	—	—	—	—	—
13	Engine runs rough/ rolling idle	—	—	—	—	—	—	—	—
14	Fast idle/runs on	—	—	—	—	—	—	—	—
15	Low idle/stalls during deceleration	—	—	—	—	—	—	—	—

Troubleshooting item		i-stop							
		Brake fluid pressure sensor (built-into DSC HU/CM)	M position switch malfunction (ATX)	Low-G (XY) sensor (built-into SAS control module)	TFT sensor (ATX)	Electric AT oil pump malfunction (ATX)	DSC HU/CM malfunction	Generator drive belt wear	Ignition switch signal (PCM terminal 2H)
			Erratic signal input	Erratic signal input	Erratic signal input				Stuck on
16	Engine stalls/quits, engine runs rough, misses, buck/jerk, hesitation/stumble, surges	—	—	—	—	—	—	—	—
17	Lack/loss of power-acceleration/cruise	—	—	—	—	—	—	—	—
18	Knocking/pinging-acceleration/cruise	—	—	—	—	—	—	—	—
19	Poor fuel economy	—	—	—	—	—	—	—	—
20	Emission compliance	—	—	—	—	—	—	—	—
21	High oil consumption/leakage	—	—	—	—	—	—	—	—
22	Cooling system concerns-overheating	—	—	—	—	—	—	—	—
23	Cooling system concerns-runs cold	—	—	—	—	—	—	—	—
24	Excessive black smoke	—	—	—	—	—	—	—	—
25	Fuel odor (in engine compartment)	—	—	—	—	—	—	—	—
26	Engine noise	—	—	—	—	—	—	—	—
27	Vibration concerns (engine)	—	—	—	—	—	—	—	—
28	A/C does not work sufficiently	—	—	—	—	—	—	—	—
29	A/C is always on or A/C compressor runs continuously	—	—	—	—	—	—	—	—
i-stop related troubleshooting item		(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)							